

MODEL QUESTION PAPER**FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY****DATA MINING-SET-I****(Maximum Marks : 75)****TIME : 3HRS****PART -A****Answer all questions in one word or one sentence****I.**

1. What is Datamining?
2. List any two application of Datamining.
3. What you meant by data integration?
4. Define ordinal attribute with an example
5. List any two data reduction methods.
6. What are support vectors?
7. Give the equation to calculate Manhattan distance
8. What you meant by frequent pattern. Give an example.
9. List any two difficulties of agglomerative clustering methods.

(9x1=9)**PART B****Answer any eight questions from the following. Each question carries 3 marks.****II.**

1. Differentiate Operational Database Systems and Data Warehouses?
2. Discuss about OLAP operations.
3. Suppose a data warehouse consists of three measures: customer, account and branch and two measures count (number of customers in the branch) and balance. Draw the schema diagram using snowflake schema.
4. Describe any two methods for attribute subset selection
5. Differentiate classification and prediction.
6. Discuss the working of backpropagation algorithm?
7. Describe confusion matrix with a suitable example.
8. Define support and confidence. With the help of an example, show how these values are calculated.
9. How to compute dissimilarity between objects described by binary variables
10. Explain Dendogram with an example.

(8x3=24)

PART C

Answer all questions from the following. Each question carries 7 marks.

III

1. Explain in detail the three tier architecture of data warehouse with a neat diagram.

OR

2. Explain the various schemas for multidimensional data model with example.
3. Explain the importance of data reduction in data warehousing. Describe the techniques used for dimensionality reduction.

OR

4. The following data is given in increasing order for the attribute age:
13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,36,40,45,46,52,70. Use smoothing by bin means to smooth these data, using bin depth of 3.
5. Suppose we have data on few individuals randomly surveyed. The data gives the responses towards interests to promotional offers made in the areas of Finance, Travel, Reading, and Health. Sex is the output attribute to be predicted. Apply Naïve Bayesian classification algorithm to classify the new instance
(Finance=No, Travel=Yes, Reading=Yes, Health=No)

Finance	Travel	Reading	Health	Sex
Yes	No	Yes	No	Male
Yes	Yes	No	No	Male
No	Yes	Yes	Yes	Female
No	Yes	No	Yes	Male
Yes	Yes	Yes	Yes	Female
No	No	Yes	No	Female
Yes	No	No	No	Male
Yes	Yes	No	No	Male
No	No	No	Yes	Female
Yes	No	No	No	Male

OR

6. Explain ID3 algorithm. Which feature selection method is used in ID3 algorithm.
7. Why linear SVM is called maximum margin Classifier. Explain with suitable figure.

OR

8. Explain Artificial Neural Network with a neat diagram. Describe the characteristics of ANN.

9. Given the data set. Use KNN algorithm to classify the new data set (T,F,4.5). Specify the k and distance measurement you use.

x_i	a_1	a_2	a_3	Class
x_1	<i>T</i>	<i>T</i>	5.0	<i>Y</i>
x_2	<i>T</i>	<i>T</i>	7.0	<i>Y</i>
x_3	<i>T</i>	<i>F</i>	8.0	<i>N</i>
x_4	<i>F</i>	<i>F</i>	3.0	<i>Y</i>
x_5	<i>F</i>	<i>T</i>	7.0	<i>N</i>
x_6	<i>F</i>	<i>T</i>	4.0	<i>N</i>
x_7	<i>F</i>	<i>F</i>	5.0	<i>N</i>
x_8	<i>T</i>	<i>F</i>	6.0	<i>Y</i>
x_9	<i>F</i>	<i>T</i>	1.0	<i>N</i>

OR

10. Discuss the various techniques used to evaluate the accuracy of a classifier.

11. Explain BIRCH algorithm.

OR

12. Consider the transaction data set given below. Set minimum support count as 2 and minimum confidence threshold as 70%. Find the frequent item set using FP Growth Algorithm.

Transaction ID	List of Item_Ids
T100	I1,I2,I5
T200	I2,I4
T300	I2,I3
T400	I1,I2,I4
T500	I1,I3
T600	I2,I3
T700	I1,I3
T800	I1,I2,I3,I5
T900	I1,I2,I3

MODEL QUESTION PAPER**FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY****DATA MINING-SET-2****(Maximum Marks : 75)****TIME : 3HRS****PART -A****Answer all questions in one word or one sentence****I.**

1. How is data warehouse different from a database?
2. What is datacube?
3. What is dimensionality reduction?
4. Define nominal attribute with example.
5. List any two techniques used to handle missing data.
6. What you meant by accuracy of a classifier.
7. Give the full form of BIRCH algorithm
8. When will you say an association rule is strong?
9. Give the equation to calculate Euclidean distance

(9x1=9)**PART B****Answer any eight questions from the following. Each question carries 3 marks.****II.**

11. Compare OLAP and OLTP?
12. Compare star and snowflake schema dimension table
13. Explain the various data warehouse models.
14. Real-world data tend to be incomplete, noisy, and inconsistent .Discuss the various approaches adopted to clean the data?
15. Explain the various issues regarding classification and prediction
16. Explain Maximum Marginal Hyperplane (MMH)?
17. Discuss the advantages of neural network.
18. What is the significance of Clustering Feature in BIRCH algorithm.
19. What are the two methods used for rule interestingness.
20. Given two objects represented by the tuples (22,1,42,10) and (20,0,36,8) .Compute the Manhattan distance between the two objects.

(8x3=24)

PART C

Answer all questions from the following. Each question carries 7 marks.

III

13. Explain various stages in knowledge discovery process with a neat diagram

OR

14. Suppose that a data warehouse for University consists of four dimensions date, spectator, location and game and two measures count and charge, where charge is the fare that a spectator pays when watching a game on the given date. Spectator may be students, adults or seniors, with each category having its own charge rate.

i) Draw a star scheme for the data warehouse.

ii) Starting with the basic cuboid [date,spectator,location,game], what specific OLAP operation should be performed in order to list the total charge paid by student spectators at GM_PLACE in 2010.

15. What is the significance of data pre-processing? Explain any four data preprocessing methods.

OR

16. Explain the various methods of Data Normalization. Normalize the following data by min-max normalization 1000,2000,3000,5000,9000 by setting min=0 and max=1

17. Consider the collection of training samples (S) in the table given below. Loan_risk is the target attribute which describes the risk associated with loan for each customer. Find the value of the following. i) Gain(S, Sex) ii) Gain(S, Credit_rating)

Cust_ID	Age	Sex	Income	Credit_rating	Loan_risk
1000	Young	F	High	Normal	Safe
1001	Young	F	High	High	Safe
1002	Middle Age	F	High	Normal	Risky
1003	Senior	F	Normal	Normal	Risky
1004	Senior	M	Low	Normal	Risky
1005	Senior	M	Low	High	Safe
1006	Middle Age	M	Low	High	Risky
1007	Young	F	Normal	Normal	Safe
1008	Young	M	Low	Normal	Risky
1009	Senior	M	Normal	Normal	Risky
1010	Young	M	Normal	High	Risky
1011	Middle Age	F	Normal	High	Risky
1012	Middle Age	M	High	Normal	Risky
1013	Senior	F	Normal	High	Safe

OR

18. Based on the following data determine the gender of a person having height 6 ft., weight 130 lbs. and foot size 8 in. (use Naive Bayes algorithm).

person	height (feet)	weight (lbs)	foot size (inches)
male	6.00	180	10
male	6.00	180	10
male	5.50	170	8
male	6.00	170	10
female	5.00	130	8
female	5.50	150	6
female	5.00	130	6
female	6.00	150	8

19. Explain Backpropagation Algorithm.

OR

20. The job market is being studied in several neighbourhoods. Let X represents the total number of jobs in a given neighbourhood. Let Y represents the number of entry level jobs in the same neighbourhood. A sample of six neighbourhood gives the following information. For a neighbourhood with 40 jobs, how many are predicted to be entry level.

x	16	33	50	28	50	25
y	2	3	6	5	9	3

21. Explain the working of K Nearest Neighbour Classifier.

OR

22. A database contains 80 records on a particular topic of which 55 are relevant to a certain investigation. A search was conducted on that topic and 50 records were retrieved. Of the 50 records retrieved, 40 were relevant. Construct the confusion matrix for the search and calculate the precision and recall scores for the search.

23. Use K-Means clustering algorithm to divide the following data into two clusters and compute the representative data points for the clusters.

x_1	1	2	2	3	4	5
x_2	1	1	3	2	3	5

OR

24. Explain Agglomerative hierarchical clustering method.

(6x7=42)

(6x7=42)